14. Difference-in-Differences

A consolidated, mathematical lecture companion

English notes based on the lecture slides *Differences in Differences* (Prof. Sanghack Lee, GSDS SNU; drawing on Matthew Blackwell and Kosuke Imai)

The central question is how repeated measurements on treated and untreated groups identify the effect of treatment on the treated when treatment is not randomized, by assuming that the untreated trend is a valid proxy for the treated group's counterfactual trend.

These notes consolidate the slides, prove the identification result, compute the Card–Krueger estimate, relate DiD to two-way fixed effects, and cover conditional and nonlinear (changes-in-changes) DiD, pre-testing, sensitivity analysis and staggered timing. Physical PDF page numbers are listed in Section D. References: Card and Krueger (1994, AER); Roth (2022, AER: Insights); Rambachan and Roth (2023, ReStud); Sant'Anna and Zhao (2020, JoE).

Contents

1 Motivation **2**

1.1 Case study: minimum wages (Card and Krueger 1994) . . . 2

2 The DiD design **2**

2.1 Setup . . . 2

2.2 The identification problem . . . 2

2.3 Three control strategies . . . 3

2.4 Parallel trends . . . 3

2.5 Estimation . . . 3

2.6 DiD and two-way fixed effects (2×2) . . . 4

3 Conditional and nonlinear DiD **4**

3.1 Conditional parallel trends . . . 4

3.2 Nonlinear DiD (changes-in-changes) . . . 4

3.3 Key takeaways . . . 5

A Pre-testing bias **5**

B Sensitivity analysis for parallel trends **5**

C Staggered treatment timing **5**

D Source coverage **6**

1 Motivation

Purpose. Introduce the two sources of variation in panel data.

So far good controls came from randomization, similar covariates (matching/weighting), an exogenous instrument (IV), or proximity to a discontinuity (RD). With repeated measurements on the same units there are two sources of variation: within-unit variation over time (before/after) and cross-sectional variation in treatment. A before–after comparison uses the treated group's own past as its counterfactual; difference-in-differences uses the control group's change as the counterfactual change of the treated group (parallel paths).

1.1 Case study: minimum wages (Card and Krueger 1994)

Does raising the minimum wage affect employment? Classical theory suggests negative effects, but minimum wages are hard to randomize. In 1992 New Jersey raised its minimum wage from \$4.25 to \$5.05 while neighboring Pennsylvania stayed at \$4.25. Employment is observed in both states before and after the increase, focusing on fast-food restaurants in eastern PA and NJ, which have similar prices, wages and products and are most likely to be affected. In the picture, average employment in NJ (treated) and PA (control) is plotted before and after; the dashed counterfactual line for NJ is parallel to PA's, and the ATT is the gap between NJ's actual and counterfactual post-period levels.

2 The DiD design

Purpose. Formalize the two-group, two-period design and identify the ATT.

2.1 Setup

Two groups, two periods. Pre-period ($t = 0$): neither group is treated. Post-period ($t = 1$): one group is treated. $G_i = 1$ for units treated at $t = 1$, $G_i = 0$ for never-treated units. Treatment status: $X_{i0} = 0$ for all; $X_{i1} = G_i$.

	Pre-period ($t = 0$)	Post-period ($t = 1$)
Treated group ($G_i = 1$)	$X_{i0} = 0$	$X_{i1} = 1$
Control group ($G_i = 0$)	$X_{i0} = 0$	$X_{i1} = 0$

Potential outcomes. $Y_{it}(x)$ is the potential outcome under treatment x at time t; the individual effect is $Y_{it}(1) - Y_{it}(0)$. Consistency: $Y_{it} = X_{it}Y_{it}(1) + (1 - X_{it})Y_{it}(0)$. Hence $Y_{i0}(0) = Y_{i0}$ for all units; $Y_{i1} = Y_{i1}(1)$ if $G_i = 1$; $Y_{i1} = Y_{i1}(0)$ if $G_i = 0$.

2.2 The identification problem

$$\text{ATT} = \mathbb{E}[Y_{i1}(1) - Y_{i1}(0) \mid G_i = 1] = \underbrace{\mathbb{E}[Y_{i1} \mid G_i = 1]}_{(a)} - \underbrace{\mathbb{E}[Y_{i1}(0) \mid G_i = 1]}_{(b)}.$$

(a) is estimable; (b) is counterfactual: what would the treated group's average have been had it remained untreated?

2.3 Three control strategies

1. **Before–after:** assume $\mathbb{E}[Y_{i1}(0) \mid G_i = 1] = \mathbb{E}[Y_{i0} \mid G_i = 1]$ (no change over time absent treatment).
2. **Cross-sectional comparison:** assume $\mathbb{E}[Y_{i1}(0) \mid G_i = 1] = \mathbb{E}[Y_{i1} \mid G_i = 0]$ (no group differences).
3. **Difference-in-differences:** assume parallel trends; then

$$\text{ATT} = (\mathbb{E}[Y_{i1} \mid G = 1] - \mathbb{E}[Y_{i0} \mid G = 1]) - (\mathbb{E}[Y_{i1} \mid G = 0] - \mathbb{E}[Y_{i0} \mid G = 0]).$$

2.4 Parallel trends

Definition 2.1 (Parallel trends)**.** $\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 1]$.

The control group's trend proxies how the treated group would have changed without treatment. Why is this weaker than other assumptions? It accounts for time-invariant unobserved differences between groups and allows for common trends over time. But it is *not invariant to nonlinear transformations*: parallel trends in Y_{it} imply non-parallel trends in $\log Y_{it}$, and vice versa.

Theorem 2.2 (Identification)**.** *Under consistency and parallel trends,*

$$\text{ATT} = (\mathbb{E}[Y_{i1} \mid G_i = 1] - \mathbb{E}[Y_{i0} \mid G_i = 1]) - (\mathbb{E}[Y_{i1} \mid G_i = 0] - \mathbb{E}[Y_{i0} \mid G_i = 0]).$$

Proof. By parallel trends, $\mathbb{E}[Y_{i1}(0) \mid G = 1] = \mathbb{E}[Y_{i0}(0) \mid G = 1] + \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G = 0]$. Consistency gives $Y_{i0}(0) = Y_{i0}$ for everyone and $Y_{i1}(0) = Y_{i1}$ for controls, so $\mathbb{E}[Y_{i1}(0) \mid G = 1] = \mathbb{E}[Y_{i0} \mid G = 1] + \mathbb{E}[Y_{i1} - Y_{i0} \mid G = 0]$. Substitute into $\text{ATT} = \mathbb{E}[Y_{i1} \mid G = 1] - \mathbb{E}[Y_{i1}(0) \mid G = 1]$. □

Threats: non-parallel trends, i.e. unmeasured time-varying confounding. *Ashenfelter's dip:* earnings often fall temporarily just before entry into a training program. The dip makes the treated group's pre-period outcome artificially low, inflating its before–after difference, so the effect is overestimated. *Falsification test:* check parallel trends in pre-treatment periods. This is necessary but not sufficient: passing does not guarantee that parallel trends hold post-treatment.

2.5 Estimation

With panel data,

$$\widehat{\text{ATT}} = \underbrace{\frac{1}{N_1}\sum_{i=1}^{N} G_i\{Y_{i1} - Y_{i0}\}}_{\text{average trend, treated}} - \underbrace{\frac{1}{N_0}\sum_{i=1}^{N}(1 - G_i)\{Y_{i1} - Y_{i0}\}}_{\text{average trend, control}},$$

with standard errors from the usual difference-in-means estimator. Equivalent regression: with $\Delta Y_i = Y_{i1} - Y_{i0}$, $\Delta Y_i = \alpha + \tau G_i + \varepsilon_i$ gives $\hat{\tau}_{\text{OLS}} = \widehat{\text{ATT}}$ (OLS on a binary regressor equals the difference in group means), convenient for adding pre-treatment covariates. DiD can also be used with repeated cross-sections.

Card and Krueger with numbers (average full-time-equivalent employment per restaurant):

	Before (Feb 1992)	After (Nov 1992)	Difference
NJ (treated)	20.44	21.03	+0.59
PA (control)	23.33	21.17	−2.16
DiD estimate			+2.76

DiD $= (21.03 - 20.44) - (21.17 - 23.33) = 0.59 - (-2.16) = 2.76$: no evidence of employment loss from the minimum-wage increase.

2.6 DiD and two-way fixed effects (2×2)

The simplified TWFE model for two groups and two periods is

$$Y_{it} = \mu + \gamma G_i + \delta t + \tau X_{it} + \varepsilon_{it},$$

with fixed effects capturing all time-invariant group differences (γG_i) and unit-invariant time effects (δt). Since $X_{it} = G_i \cdot t$, the four cell means are μ, $\mu + \gamma$, $\mu + \delta$, $\mu + \gamma + \delta + \tau$, and the coefficient τ equals the DiD estimate.

3 Conditional and nonlinear DiD

Purpose. Weaken parallel trends with covariates, and replace mean trends by rank preservation.

3.1 Conditional parallel trends

Unconditional parallel trends may fail if groups differ in observed covariates X_i. *Conditional parallel trends*: for all x,

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 1, X_i = x] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0, X_i = x].$$

This is weaker: trends need only be parallel within covariate strata (states with different demographics may trend differently, but states with similar demographics should evolve similarly). Estimators:

- **Outcome regression:** include X_i in the (TWFE) model.
- **IPW-DiD:** reweight control units by $\hat{e}(X_i)/(1 - \hat{e}(X_i))$ to match the treated group's covariate distribution.
- **Doubly robust DiD:** combine both; consistent if either the outcome model or the propensity model is correct (Sant'Anna and Zhao 2020; Callaway and Sant'Anna 2021), with the same logic as DR estimation in Lecture 10-2.

3.2 Nonlinear DiD (changes-in-changes)

Standard DiD's parallel trends in levels are not preserved under log, $\sqrt{\cdot}$, etc. Key insight: assume parallel trends in *ranks* (quantiles) instead of levels: absent treatment, a unit at the q-th quantile of its group in period 0 remains at the q-th quantile in period 1. Its value changes, but its rank is preserved. This rank-preservation assumption is invariant to monotone transformations (the q-th quantile of Y maps to the q-th quantile of $\log Y$).

Formalization (Athey and Imbens 2006). Let $F_{gt}(y) = P(Y_{it}(0) \leq y \mid G_i = g)$. Observed: F_{00}, F_{01}, F_{10} (untreated outcomes), and the treated post-period CDF $F_{11}^{(1)}$ of $Y_{i1}(1)$. Goal: the counterfactual F_{11}. Quantile treatment effect:

$$\tau(q) = \left(F_{11}^{(1)}\right)^{-1}(q) - F_{11}^{-1}(q).$$

Define the period-0-to-1 value shift at each rank: $h_0(y) = F_{01}^{-1}(F_{00}(y))$ for controls and $h_1(y) = F_{11}^{-1}(F_{10}(y))$ for the treated. *Identification assumption:* $h_0 = h_1$, i.e. the value shift at each rank is the same in both groups. Then $F_{11}^{-1}(F_{10}(y)) = F_{01}^{-1}(F_{00}(y))$, and solving,

$$F_{11}(y) = F_{10}(F_{00}^{-1}(F_{01}(y))).$$

(Take $y' = F_{01}^{-1}(F_{00}(y''))$; the assumption says $F_{11}(y') = F_{10}(y'')$ with $y'' = F_{00}^{-1}(F_{01}(y'))$.)

3.3 Key takeaways

- Parallel trends are the core of DiD identification.
- DiD handles time-invariant unobserved confounding but is vulnerable to time-varying confounders (Ashenfelter's dip).
- Conditional parallel trends with covariates lead to IPW-DiD and doubly robust DiD.
- Pre-trend tests are useful but do not validate parallel trends; sensitivity analysis is preferred.
- Linear DiD is not invariant to nonlinear transformations; nonlinear DiD needs distributional assumptions.
- TWFE gives the same result in the 2×2 case but needs care with staggered timing.

A Pre-testing bias

Common practice: run a pre-trend test and proceed with DiD only if it is insignificant. Problem: conditioning on passing biases the DiD estimate. Tests have low power against small pre-trends and often fail to reject; "passing" selects samples in which pre-trends happen to look flat. The same chance variation that hid the pre-trend persists into the post-period and distorts the estimate. Result: under-rejection of the pre-trend test leads to over-rejection of "no treatment effect". A passed pre-trend test gives comfort, not validation (Roth 2022).

B Sensitivity analysis for parallel trends

Parallel trends are untestable; how robust are the estimates? Allow bounded violations: let M bound the deviation

$$\left|\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 1] - \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0]\right| \leq M.$$

For each M construct a confidence set for the ATT; as M grows the set widens. Report the *breakdown value*, the smallest M at which the set first includes 0 (the finding is no longer robust). One can also extrapolate from pre-treatment trend differences to choose plausible bounds (Rambachan and Roth 2023).

C Staggered treatment timing

Units often enter treatment at different times. Standard TWFE with staggered timing can be biased. The Goodman-Bacon decomposition shows TWFE is a weighted average of all possible 2×2 DiD comparisons: early-treated vs. never-treated (good), late-treated vs. never-treated (good), and late-treated vs. early-treated (dangerous: the early-treated serve as controls). If

effects change over time, early-treated units are not valid controls for the late-treated, since their outcomes already reflect treatment; using them attenuates or even reverses the estimate.

Modern solutions. Callaway and Sant'Anna (2021): compute group-time average treatment effects using only never-treated or not-yet-treated units as controls. Event-study plots: plot effects by time relative to treatment onset, $t - t^*$, allowing visual inspection of pre-trends and dynamic effects. (See the lecture on staggered DiD.)

D Source coverage

The source deck has 44 physical pages (25 numbered slides).

Topic	Physical pages
References, Card–Krueger figure, motivation, variations	1–9
Design, potential outcomes, identification, strategies, parallel trends	10–22
Estimation, Card–Krueger numbers, TWFE	23–27
Conditional PT, estimators, nonlinear DiD, takeaways	28–37
Appendix: pre-testing, sensitivity, staggered timing	38–44